

GRENADIER ISLAND, ON, Canada

WMO: 712810

Lat: 44.420N

Lon: 75.850W

Elev: 82

StdP: 100.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.1	(c) -19.0	(d) -26.8	(e) 0.3	(f) -21.0	(g) -23.6	(h) 0.5	(i) -18.1	(j) 11.0	(k) -2.3	(l) 9.2	(m) -0.3	(n) 1.8	(o) 20	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.5	23.4	28.1	22.5	26.8	21.7	24.6	28.1	23.5	26.8	22.5	25.4	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.5	26.8	22.5	17.4	25.5	21.6	16.4	24.4	75.1	28.3	70.8	26.9	66.9	25.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 6.8	(c) 5.6	(e) -25.6	(f) 31.8	(g) 3.6	(h) 1.6	(i) -28.2	(j) 32.9	(k) -30.3	(l) 33.9	(m) -32.4	(n) 34.7	(o) -35.0	(p) 35.9		
(a) 8.3	(b) 6.8	(c) 5.6	(e) -25.8	(f) 26.3	(g) 3.6	(h) 1.3	(i) -28.4	(j) 27.2	(k) -30.5	(l) 27.9	(m) -32.5	(n) 28.6	(o) -35.2	(p) 29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-7.1	-5.5	-0.1	6.8	13.2	18.5	21.1	20.7	16.7	10.0	3.8	-2.6
	DBStd	10.85	7.19	5.93	5.75	4.43	3.72	3.25	2.55	2.79	3.67	4.21	4.76	5.56
	HDD10.0	2039	529	434	314	114	12	0	0	0	1	52	192	391
	HDD18.3	4006	787	668	570	347	164	38	6	10	72	258	436	649
	CDD10.0	1323	0	0	3	18	110	254	343	331	203	54	6	1
	CDD18.3	245	0	0	0	0	4	42	90	83	24	2	0	0
	CDH23.3	1538	0	0	1	4	43	279	564	501	140	7	0	0
	CDH26.7	317	0	0	0	0	4	55	117	111	31	0	0	0
Wind	WSAvg	2.4	2.7	2.5	2.5	2.6	2.2	1.9	1.9	1.9	2.2	2.4	2.8	2.7
Precipitation	PrecAvg	982	74	61	65	79	83	82	92	79	92	104	92	84
	PrecMax	1226	154	107	121	144	146	175	198	118	173	182	144	143
	PrecMin	782	26	25	28	10	33	18	33	24	49	19	42	37
	PrecStd	106	32	18	25	36	34	39	37	31	31	46	26	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	9.1	19.8	23.1	26.9	30.5	30.9	31.2	30.0	24.2	17.1	12.2
		MCWB	9.7	5.0	14.3	16.1	20.1	23.4	24.8	24.0	22.5	19.1	13.8	10.1
	2%	DB	7.8	6.5	13.3	19.3	24.3	28.2	29.3	29.3	26.9	20.9	14.8	8.9
		MCWB	6.9	4.6	9.2	12.0	17.2	22.4	23.8	23.0	21.7	17.2	12.3	7.5
	5%	DB	4.6	4.6	10.3	16.3	22.3	26.5	28.0	27.8	24.6	18.6	12.8	6.9
		MCWB	3.7	2.7	6.9	10.4	15.9	21.3	22.6	22.2	20.0	15.4	10.8	5.6
	10%	DB	2.2	2.7	7.6	13.9	20.0	24.7	26.6	26.2	22.7	16.5	10.6	4.6
		MCWB	1.4	1.4	4.8	9.0	14.9	20.0	21.9	21.1	18.8	13.8	8.7	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	6.6	14.5	17.0	21.6	25.0	26.3	25.7	23.7	20.4	14.7	11.0
		MCDB	10.6	7.6	19.0	21.5	25.2	29.4	29.7	29.3	28.0	22.7	16.0	12.3
	2%	WB	6.8	4.8	10.0	13.6	19.0	23.3	24.7	24.4	22.1	17.6	12.9	7.8
		MCDB	7.7	6.3	12.4	17.2	22.3	26.9	28.1	27.8	25.7	19.7	14.4	8.7
	5%	WB	3.7	3.0	7.4	11.6	17.3	21.9	23.7	23.2	20.9	15.9	11.2	5.6
		MCDB	4.6	4.3	9.9	14.7	20.5	25.2	26.8	26.0	23.7	18.0	12.6	6.6
	10%	WB	1.4	1.5	5.2	9.6	15.8	20.7	22.6	22.2	19.7	14.2	8.9	3.5
		MCDB	2.1	2.6	7.1	13.3	18.7	23.8	25.4	24.8	21.9	16.1	10.3	4.5
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.0	9.2	9.7	10.0	9.9	9.5	9.7	9.6	8.1	7.0	6.8
		MCDBR	8.5	9.1	12.5	13.8	13.3	12.0	10.8	11.3	11.7	10.9	10.0	8.2
		MCWBR	8.1	7.5	8.5	7.9	7.2	6.8	6.1	6.1	6.7	7.3	7.8	7.2
	5% WB	MCDBR	8.6	8.8	11.8	12.3	11.5	10.9	9.8	9.9	10.5	10.1	9.4	8.2
		MCWBR	8.4	7.8	8.8	8.5	7.4	6.8	6.2	5.9	6.5	7.3	8.0	7.6
Clear-Sky Solar Irradiance	taub	0.290	0.295	0.324	0.354	0.383	0.417	0.429	0.418	0.369	0.349	0.310	0.285	
	taud	2.374	2.362	2.417	2.394	2.357	2.278	2.251	2.282	2.407	2.456	2.509	2.455	
	Ebn at Noon	830	896	914	908	887	854	839	834	853	823	809	806	
	Edh at Noon	76	92	101	112	120	131	133	125	101	85	68	64	
All-Sky Solar Radiation	RadAvg	1.34	2.23	3.55	4.41	5.29	5.66	5.79	5.10	4.06	2.41	1.51	1.03	
	RadStd	0.13	0.23	0.29	0.53	0.55	0.45	0.50	0.24	0.34	0.23	0.17	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.31	+1.46	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page